Unit 2 - Algebra

Section 2.0: Algebra Basics

Concept 1: Variables

In statistics, we defined the notion of a ***variable***.

Variable (Statistics) – A property of the individual (that varies from one individual to another).

In algebra, we are also interested in variables. We use the term in several related ways.

A **variable** is a real-world _____ whose value is unknown or not specified.

Example 1: Let x represent the number of seats on a bus.

A **variable** is the _____ that we use to represent some quantity.

Example 2: The equation $y = 2x - 1$ contains _____ variables (_____ and _____), each of which appears _____ time.

Example 3: The equation $3x - 4 = 2x + 3$ contains _____ variable (_____) which appears _____ times.

Example 4: The formula $I = Prt$ (the *simple interest formula*) contains _____ variables, which represent:

I - _____

P - _____

r - _____

t - _____

Notice that, when a variable has a real-world context, we need two pieces of information to say what it represents:

- The _____ that the variable represents
- What _____ that thing will be measured in

This is because when we do algebra, we only work with _____, but in real life, the meaning of a number depends on its _____.

Example 5: In the simple interest formula, $t = 4$ means 4 _____.

If we wanted to represent 4 months, we had to use $t =$ _____.

Concept 2: Expressions

An ***expression*** is the most basic unit we work with in algebra.

An **expression** is a sequence of symbols consisting of numbers, variables, and symbols for mathematical operations which represents a number.

An expression which contains one or more variables is a **variable expression**. An expression which contains only numbers and symbols for operations is a **numeric expression** or **arithmetic expression**.

Example 6:

$3x - 1$ is a _____ expression with _____ variable.

$(-2)^3 + 11$ is a _____ expression.

When our variables are defined in context, we can interpret our expressions in context.

Example 7: Let x represent the number of cookies in a package.

$3x$ represents _____

_____

$2x + 5$ represents _____

_____

$\frac{x}{3}$ represents _____

_____

We can represent the number of cookies in 4 packages if 3 cookies are removed by writing _____

When we have a complicated expression, we may want to write a simpler ***equivalent***. This process is called simplifying.

Two expressions are **equivalent** if they have the same value, no matter what number the variable represents.

When we **simplify** an expression, we write the simplest possible equivalent expression. It may be necessary to define the "simplest possible" form for more complicated types of expressions.

Example 8: The expression $2x + 3x$ is equivalent to the expression $5x$. (We can confirm that by seeing what number this represents for various values.

x	$2x + 3x$	$5x$
0		
5		
-3		
12		

Since it's easier to write and evaluate, we say that $5x$ is _____ than $2x + 3x$. If we rewrite $2x + 3x$ as $5x$, we can say we have *simplified* the expression.

We have the following moves available to us to simplify expressions.

1. Do arithmetic operations to combine numbers.
   Example 9:
   - We can rewrite $3 \cdot 2x + 3 + 5$ as _____

     Think of this as _____

     _____
2. Use the ***associative property*** to eliminate parentheses if we are only using addition or only using multiplication.
   Example 10:
   - We can rewrite $(x + 4) + 1$ as _____

     We can think of this as _____

     _____
3. We can use the ***distributive property*** to eliminate parentheses when multiplication and addition are mixed.

   Example 11:
   - We can rewrite $3(x + 1)$ as _____

     Think of this as _____

     _____
4. We can ***combine like terms*** to decrease the number of variable terms.
   Examples 12:
   - We can rewrite $3x + 5 + 2x$ as _____

     Think of this as _____

     _____
   - We can rewrite $5x - 2 - 3x + 4$ as _____

     Think of this as _____

     _____

When we ***evaluate*** (work out the value of) expressions, we must be aware of order of operations. It's possible to over-think order of operations, but the key ideas are:

1. Exponents come before multiplication.
2. Multiplication comes before addition.
3. If we want to violate this order, we use grouping symbols (parentheses, fraction bars).

Notice that we didn't mention _____ or _____.
That's because these aren't "real" operations.

- Division is really just _____ : $6 \div 3$ is $6 \cdot$ _____
- Subtraction is really just _____ : $5 - 2$ is $5 +$ _____

In algebra, we almost never write _____ without using a fraction bar.

Example 13: Evaluate $3 \cdot (-2)^2 + 5 \cdot (-2) - 7$

If we have a _____ expression, we must know what number the variable(s) represent before we can evaluate.

Example 14: Evaluate $3x - 7y + 4$ when $x = 5$ and $y = 2$.

Concept 3: Equations

The most common object to work with in an algebra class is an ***equation***.

An **equation** is a statement that two (or more) expressions are equal, that is, they represent the same number.

Example 15:

1. The equation $2 + 3 = 5$ says that the expressions _____ and _____ are equal.
2. The equation $2x + 3 = 11$ says that the expressions _____ and _____ are equal.
3. The equation $y = \frac{1}{2}x - 4$ says that the expressions _____ and _____ are equal.
4. The equation $\frac{3}{4} = 0.75 = 75\%$ says that the expressions _____, _____, and _____ are all equal to each other.
5. The equation $I = Prt$ says that the expressions _____ and _____ are equal.

As the above examples show, we ***use*** equations in several ways.

1. To show the _____ of arithmetic.
2. To place a _____ on an unknown number.
3. To show a _____ between two unknown numbers.
4. To show that a _____ can be expressed in different ways.
5. To show a _____ between real-world quantities.

The simplest equations in algebra are ***linear equations*** in ***one variable***.

An expression or equation is **linear** if the only operations that appear are addition (and subtraction) or multiplication (and division) by *numbers*. If variables are multiplied together, variables are divided by, powers of variables are taken, or variables appear in exponents, the expression or equation is not linear, or **non-linear**.

The **number of variables** in an expression or equation is the number of *different* variable symbols that appear. The *same* variable can appear more than once without changing the number of variables.

Examples:

- $2x + 3$ is a _____
- $4x - 5y$ is a _____
- $3x + 1 = 2x - 7$ is a _____
- $4x - 3y = 12$ is a _____
- $x^2 + 1$ is a _____
- $y = 0.2x^2 - 7x + 4$ is a _____
- $2^x - 4$ is a _____

One of the things we will want to do, especially with equations in one variable, is to ***solve*** them.

A **solution** to an equation is value(s) of the variable(s) that make the equation true, that is, makes the expressions equal.

To **solve** an equation means to find all possible solutions.

Example 16:

- $x = 4$ is a solution to the equation $3x - 5 = 7$ because _____

  _____
- $x = 1$ is a solution to the equation $2x^2 - 5x + 4 = 1$ because _____

  _____
- $x = 3$ is a solution to the equation $3^x = 5x + 12$ because _____

  _____

  _____

We will often use _____ to solve equations.

Concept 4 – Solving a Linear Equation in One Variable

We have three basic moves available to us to solve equations. (There are more advanced moves that you would learn if you took an advanced algebra class.)

1. Simplify the _____ on each side of the equation.
2. Add or subtract the same _____ on both sides of the equation.
3. Multiply or divide by the same _____ (but not _____) on both sides of the equation.

When we use any of these moves, we get an ***equivalent equation***.

Two equations are **equivalent** if they have exactly the same solutions.

Example 17: The equations $2x + 3 = 7$ and $3x - 1 = 5$ are equivalent, because both have $x = 2$ as their only solution.

Here's how we'll apply these three moves to solve any equation.

One-step equations

A one-step equation is one in which the variable appears only once, and only one operation is applied to it. To solve, we'll do the opposite of that operation.

Example 18: Solve $x + 5 = 11$.

Here, _____ is being _____ to x. The opposite of _____ is _____.

Example 19: Solve $\frac{2}{3}x = 10$

Here, x is being _____ by _____. The opposite of _____ is _____.

Two-step equations

A two-step equation is one in which the variable appears only once, but both multiplication and addition (or subtraction) are applied to it, without parentheses. To solve, we'll reverse first the _____ then the _____.

Example 20: Solve $4x - 7 = 9$

First, we _____ on both sides.

Then, we _____ on both sides.

The solution is $x =$ _____

Equations with variables on both sides

When the variable appears on both sides of the equation, we first _____ or _____ to eliminate one of the variable terms.

Example 21: Solve $5x + 4 = 2x - 5$

First, we subtract _____ on both side, and _____ .

Now we have a _____ !

Next, we _____ on both sides.

Finally, we _____ on both sides.

Equations that don't look like any of those

If we have a linear equation, it's always of one of these types. But we may need to simplify or otherwise rewrite the equation to be able to see it.

Example 22: $29 - x = 11$

Example 23: $12 + 5(x - 1) + 4 = 26$

Concept 5 – Finding Solutions to an Equation in Two Variables

An equation in two variables needs _____ numbers to have a solution: one number for each _____.

In general, equations in two variables will have _____ solutions.

To find a solution to an equation in two variables:

1. Choose a _____ for one _____ and plug it in.
2. You now have an equation in _____ variable!
3. _____ the new equation.
4. Answer as an *ordered pair*.

Example 24: Find *a* solution to the equation $y = 2x - 1$

We see that this will be easier if we pick a value for _____ and find the value of _____.

One solution is _____.

Note: in the ordered pair, the first number represents the value of _____ and the second the value of _____. If our variables are *x* and *y*, they will *always* be in this order.

To make the order clear, we could write _____

Example 25: Find three different solutions to the equation $y = \frac{2}{3}x + 5$, at least one by choosing a value of *y*.

First solution: choose _____.

Second solution: choose _____.

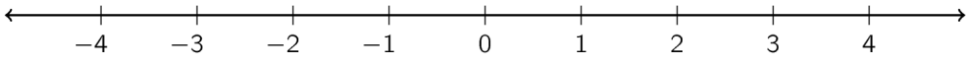
Third solution: choose _____.

The three solutions we found are: _____

Concept 6 – Coordinate Planes

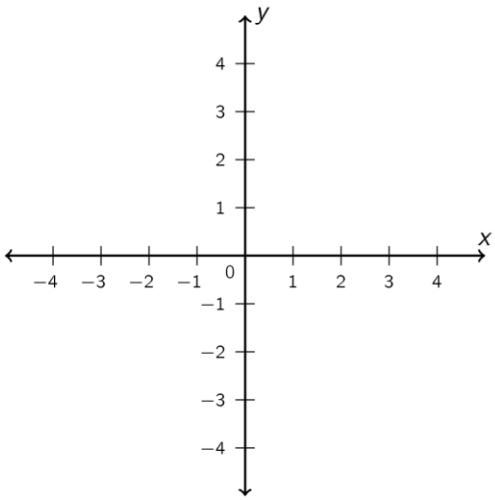
We can represent a single variable as a point on a number line.

Example 26: Graph $x = 2$ on the number line below.

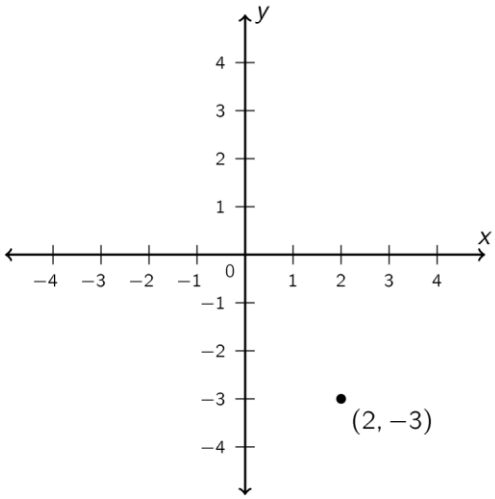


Graphing gets more interesting when we are interested in two variables.

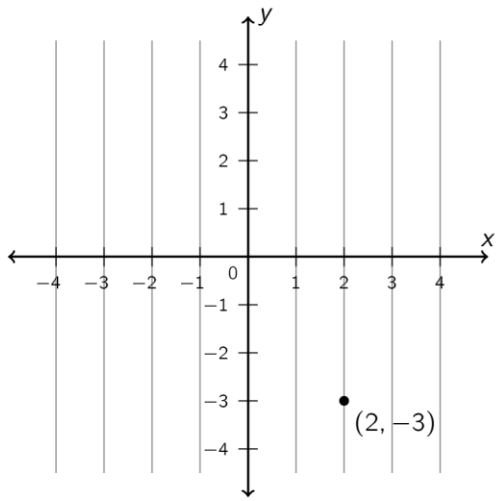
To graph two variables, we draw two number lines at right angles to each other, called *axes*. They are labeled with numbers and with variable names.



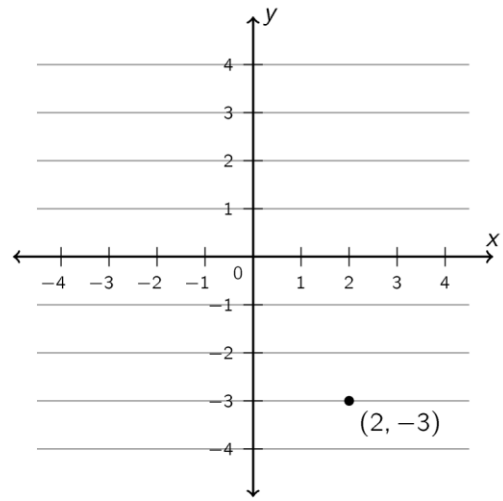
This is called a *coordinate plane*. Each point on such a plane represents a pair of values. For example, the point marked below represents the pair $x = 2$, $y = -3$. We can also write this as the ordered pair $(-2,3)$. When we're graphing, the first number always represents the number shown on the horizontal axis, the second on the vertical axis.



In order to make a coordinate plane easier to read, we often include a grid. In our grid, each vertical line represents points with the same x-coordinate (that is, at the same horizontal distance), and each horizontal line represents points with the same y-coordinate (that is, at the same horizontal distance). Notice which lines the point $(-2,3)$ lands on.



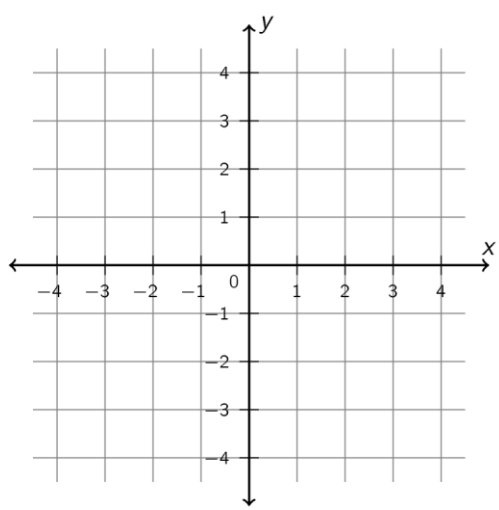
Vertical line labeled with the ____ coordinate



Horizontal line labeled with the ____ coordinate

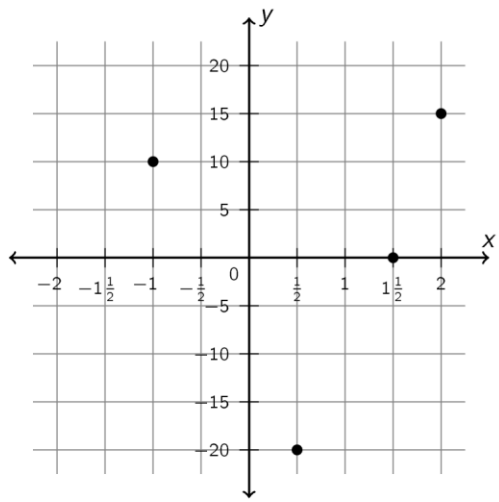
If we want to "plot" a point on a coordinate plane, then, we use the labels on the horizontal and vertical grid lines.

Example 27: Plot the points $(4,1)$, $(-2,3)$, and $(0,-2)$ on the axes below.



We don't need to count by 1s when we label axes. We can give fractional labels, or skip count. It is important to look at the numbers used to label the axes, not just to count spaces.

Example 28: Label the points on the graph below with their coordinates.

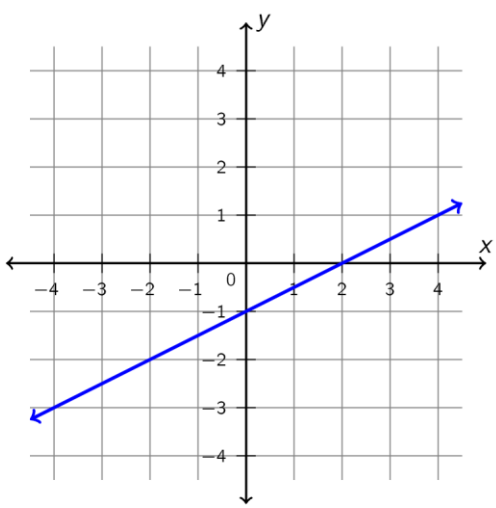


Note: While the *x*-axis may have different spacing than the *y*-axis, each axis should have the same spacing all along its length.

Concept 7 – Graph of a Linear Equation

We can use a coordinate plane to graph an equation in two variables. The graph consists of all the pairs of values that are solutions to the equation.

Example 29: The graph below is the graph of the equation $y = \frac{1}{2}x - 1$. Choose some points on the graph and observe that their coordinates are solutions to the equation.

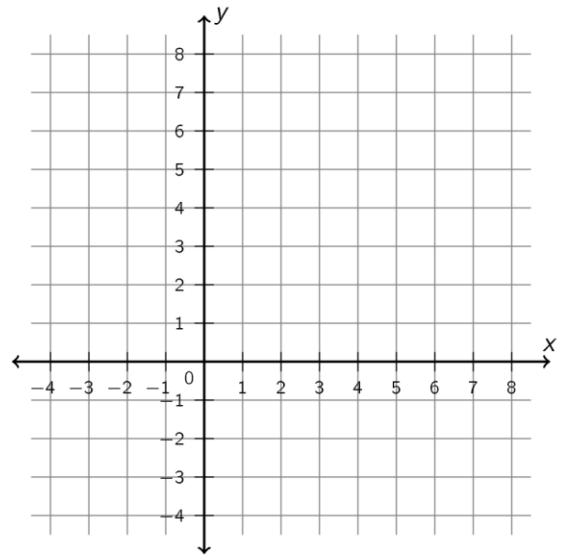


From geometry, we know that it takes two points to "fix" a line: that is, if we know two points, there is only one line that we can draw through them. That means that if we want to draw the graph of an equation in two variables, we need to know two points on the line.

Example 30: Find two solutions to the equation $3x + 2y = 12$. Use them to sketch the graph of this equation.

First solution:

Second solution:



This method can be difficult, because we often get fractions when we try to find solutions to an equation in two variables. When a linear equation in two variables is given in a certain specialized form, we have a simpler method of graphing the equation.

Slope-Intercept Form of an Equation

A linear equation in two variables is in **slope-intercept form** if it is written as

$$\boldsymbol{y = mx + b}$$

where *m* and *b* are numbers. The number *m* is called the **slope** of the line, and the number *b* is called the **intercept**.

Example 31: The equation $y = 2x - 3$ is in slope-intercept form.

a. Identify the slope and the intercept from the equation.

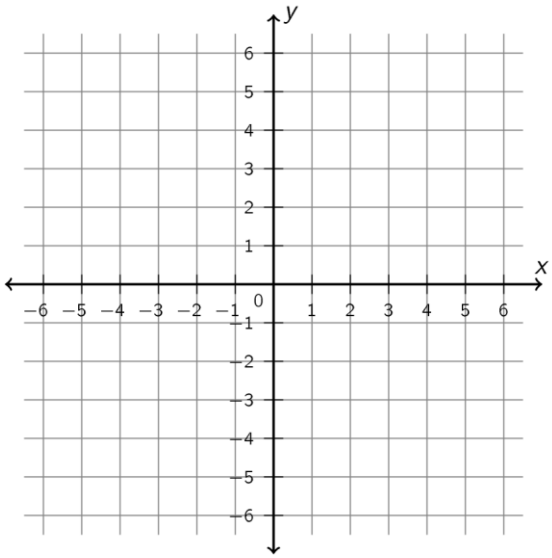
Slope: _____

Intercept: _____

b. Fill in the table below with solutions of the equation.

x	y	**Work**
−2		
−1		
0		
1		
2		

c. Using two of the solutions in the table, draw a graph of the equation.



d. Where do we see the slope and the intercept in the graph?

The slope is _____

_____

The intercept is _____

Example 32: The equation $y = -\frac{3}{2}x + 1$ is in slope-intercept form.

a. Identify the slope and the intercept from the equation.

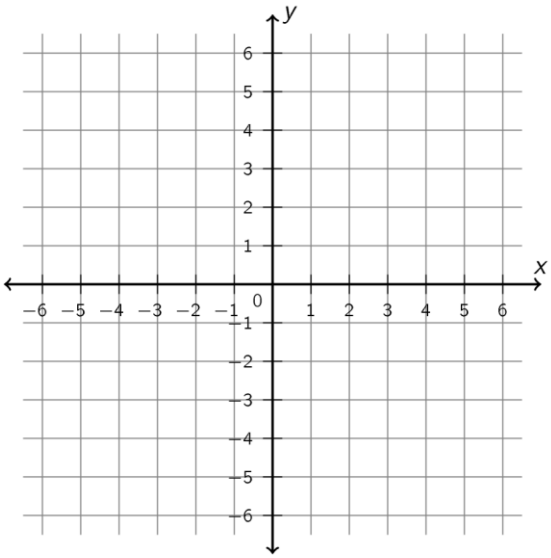
Slope: _____

Intercept: _____

b. Fill in the table below with solutions of the equation.

x	y	Work
−4		
−3		
−2		
−1		
0		
1		
2		
3		
4		

c. Using two of the solutions in the table, draw a graph of the equation.



d. Where do we see the slope and the intercept in the graph?

The slope is _____

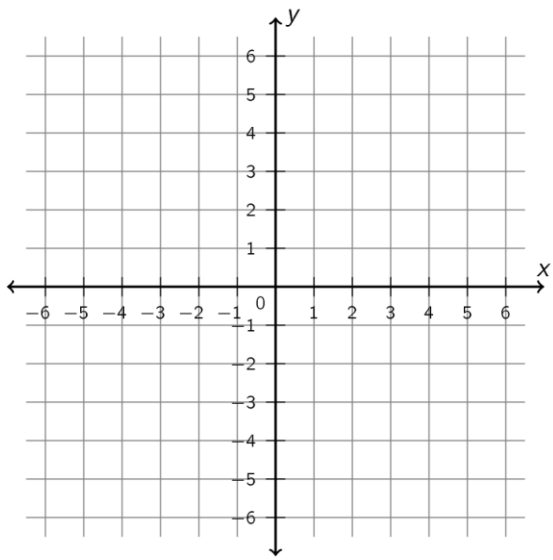
_____

The intercept is _____

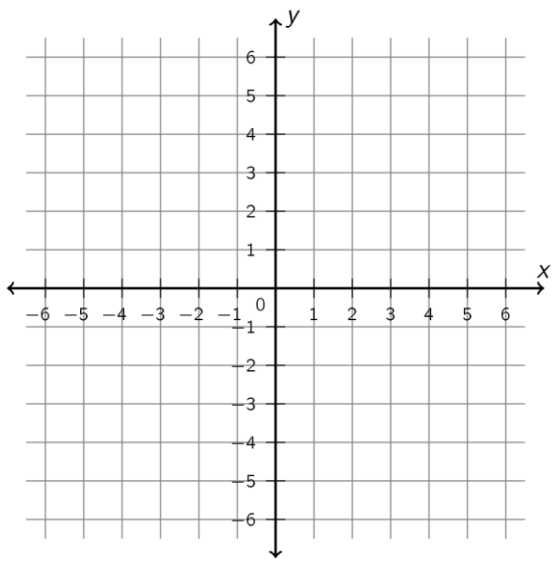
This gives us a method of drawing the graph of an equation given in slope-intercept form *without actually finding solutions to the equation*. (Notice that this method is just a shortcut to using the tables above!)

1. Find the *intercept* on the *y*-axis. (This point is the *y*-intercept of the line.)
2. Find the *denominator* of the slope. If the slope isn't a fraction, its denominator is 1. Starting from the *y*-intercept, count that many spaces to the right.
3. Find the *numerator* of the slope. If the slope is positive, count that many spaces up. If the slope is negative, count that many spaces down.

Example 33: Use the shortcut to sketch the graph of $y = \frac{3}{4}x - 2$

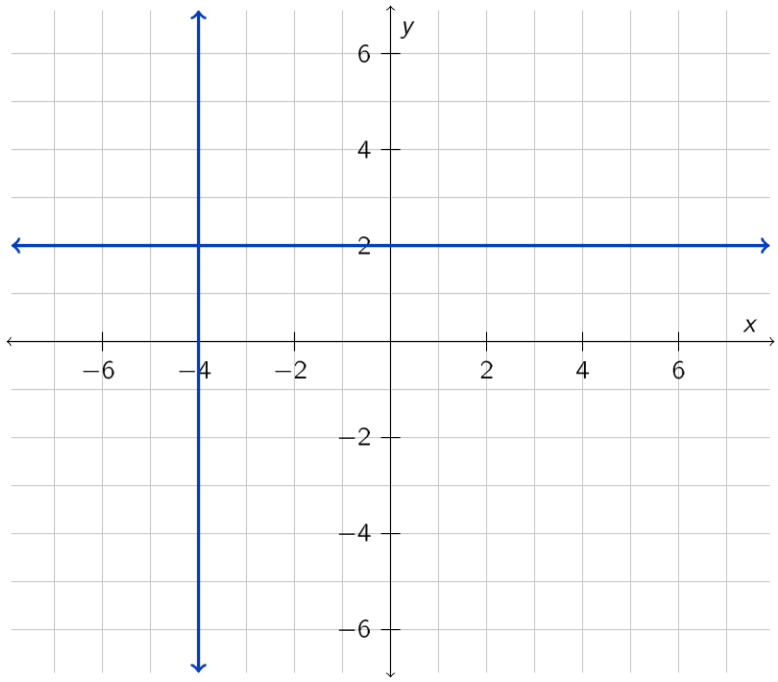


Example 34: Use the shortcut to sketch the graph of $y = -2x + 5$



The kinds of lines we have graphed so far are called "oblique" lines. We can also graph a horizontal or a vertical line.

The graph below shows a horizontal line and a vertical line.



- The points on the *vertical* line all have the same _____

  The equation of this line is _____
- The points on the *horizontal* line all have the same _____

  The equation of this line is _____

EQUATIONS FOR HORIZONTAL AND VERTICAL LINES

The graph of $\boldsymbol{y = b}$ where b is a constant is a _____ line.

The graph of $\boldsymbol{x = a}$ where a is a constant is a _____ line.